

Solutions to Problem Set 4, Second Half

180.301 Microeconomic Theory - Spring 2026

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1. In the following, I think of α as a positive number, which means that the insurer's constraint is $-(1-p)\alpha + p\beta = 0$. This does not change anything substantive; it's just for convenience. You can, of course, stick to the problem set's description and think of α as a negative number; it changes nothing except the sign.

To use the notations in class, this problem has $w_0 = 1000$, $x = 400$, $\lambda = 1.1$.

(a) $-(1-p)\alpha + \lambda p\beta = 0 \Rightarrow -0.95\alpha + 0.055\beta = 0 \Rightarrow \beta = \frac{190}{11}\alpha$.

The expected utility of the DM is: $EU = (1-p)\ln(w_G) + p\ln(w_B) = 0.95\ln(1000-\alpha) + 0.05\ln(600+\beta) = 0.95\ln(1000-\alpha) + 0.05\ln(600+\frac{190}{11}\alpha)$.

We want to maximize this. So, we take the derivative of EU with regard to α :

$$\frac{dEU}{d\alpha} = -\frac{0.95}{1000-\alpha} + \frac{0.05 \times 190/11}{600+\frac{190}{11}\alpha} = 0.$$

Solving, we get $\alpha^* = 17$, $\beta^* = \frac{190}{11}\alpha^* = \frac{3230}{11} \approx 293.64$.

- (b) The minimum deductible is the loss minus gross coverage of the insurance: $x - (\alpha + \beta) = 400 - (17 + 293.64) = 89.36$.

- (c) We just replace the 600 with 500 in question (a):

$$EU = 0.95\ln(1000-\alpha) + 0.05\ln(500+\frac{190}{11}\alpha).$$

$$\frac{dEU}{d\alpha} = -\frac{0.95}{1000-\alpha} + \frac{0.05 \times 190/11}{500+\frac{190}{11}\alpha} = 0.$$

Solving, we get $\alpha^* = 22.5$, $\beta^* = \frac{4275}{11} \approx 388.64$. The optimal level of coverage β^* increases.

- (d) Under fair insurance, $\lambda = 1$, we have $-(1-p)\alpha + p\beta = 0 \Rightarrow \beta = \frac{1-p}{p}\alpha$

We know that, under fair insurance, the DM will choose full insurance (see the slides from Apr 24th for proof):

$$w_G = w_B \Rightarrow 1000 - \alpha = 600 + \beta \Rightarrow 1000 - \alpha = 600 + \frac{1-p}{p}\alpha.$$

Solving, we get $\alpha^* = 400p$, $\beta^* = 400(1-p)$.

So, if p increases, the optimal insurance premium α^* will increase, and the optimal coverage β^* will decrease.

- (e) $v(w) = \ln(w^2) = 2\ln w = 2u(w)$. So, the new utility function $v(w)$ is just an increasing linear transformation of $u(w)$; the two utility functions represent the same preference (see pages 14-15 of Prof. Daley's Slides 1). So, the decision maker with $v(w)$ will choose the same optimal insurance as the one with $u(w)$.
2. (a) An equilibrium in this problem is a set C of insurance policies offered such that:
- Every policy $(\alpha, \beta) \in C$ is (weakly) profitable
 - For every other policy $(\alpha', \beta') \notin C$, if it is to be offered along with existing policies in C , then (α', β') is not profitable.
- (b) The reasoning of the proof is the same as in class where there were only two types. In this problem with three types, no two types can be pooled together and offered a single insurance contract.

Figure 1 illustrates why any two types cannot be pooled together. The “Pop” line is the average of the two types.

Any policy below the Pop line cannot be the equilibrium contract that attracts both H and L types, because a new contract slightly to its northeast can be offered, which is better for the insureds and also profitable for the insurer.

Any policy above the Pop line cannot be the equilibrium contract that attracts both H and L types, because the insurer will make a loss with this contract.

Any policy on the Pop line cannot be the equilibrium contract that attracts both H and L types. Consider such a policy A . We draw both type's indifference curves that pass through A . Then we can offer a new policy in the “southeastern” region between the two ICs (the yellow region) that attracts the low-risk type and is profitable.

So, no two types can be pooled together in equilibrium. Using the same reasoning, the three types also cannot be all pooled together in equilibrium. So, there is no pooling equilibrium.

- (c) Figure 2 illustrates the separating equilibrium. The reasoning is essentially the same as the problem in class with only two types.

The H type will get full insurance.

Draw H's indifference curve that passes through the full insurance point. Then, M's equilibrium is at the intersection of H's indifference curve and M's fair insurance line.

Draw M's indifference curve that passes through M's equilibrium point. Then, L's equilibrium is at the intersection of M's indifference curve and L's fair insurance line.

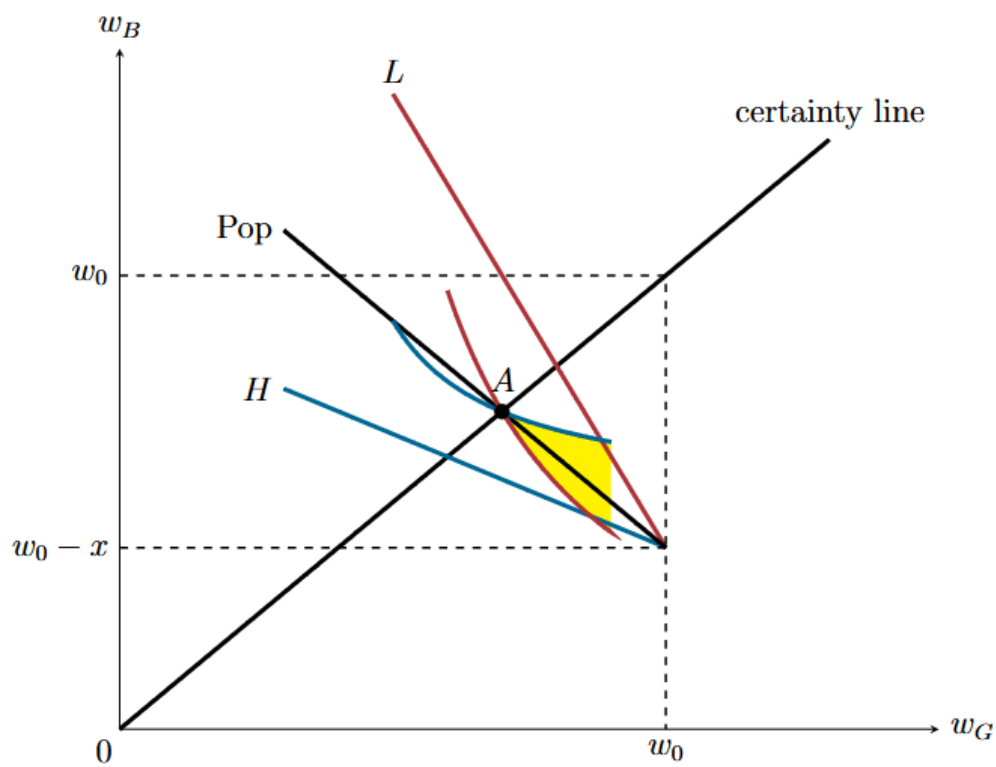


Figure 1: No pooling equilibrium with two types

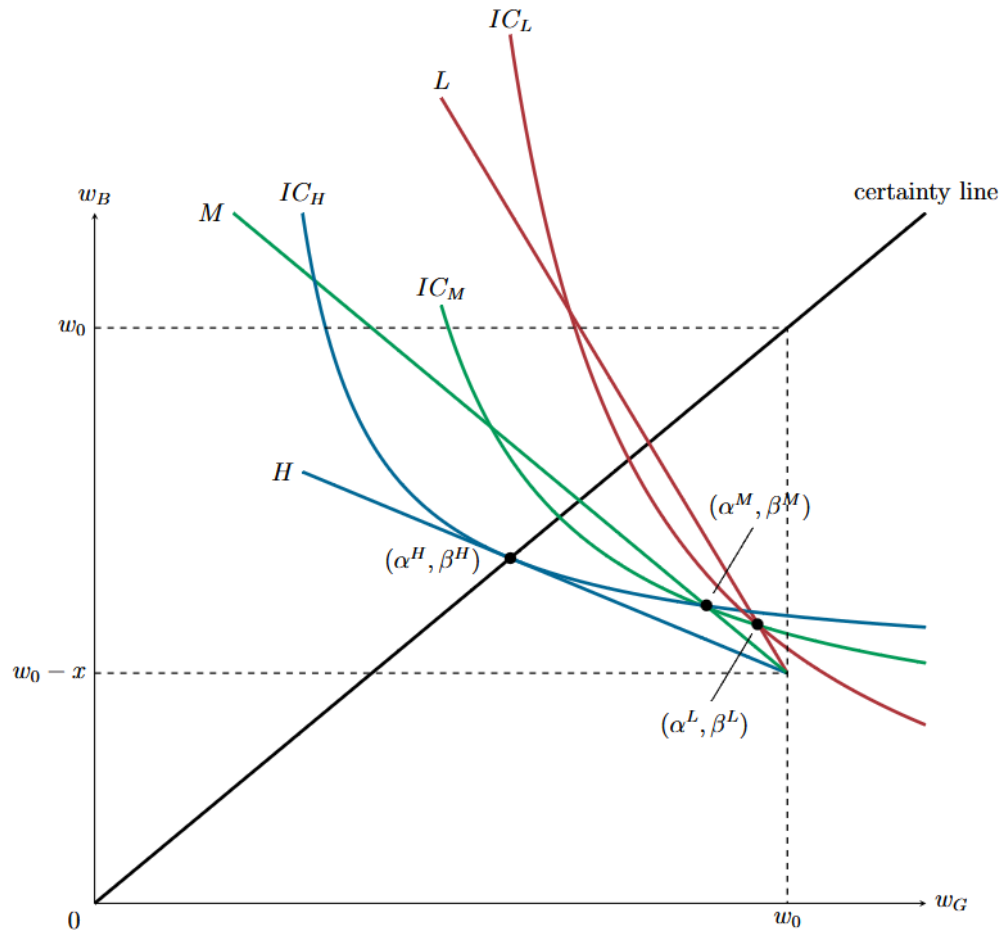


Figure 2: The separating equilibrium